DETAILED RESPONSE

TO THE EDITOR AND REVIEWERS

JOURNAL	Discover Sustainability
MANUSCRIPT ID	96bada1b-f920-471d-9cde-91916e09cc19
REVISION ROUND	Major revision of version 2.0
REVISED TITLE	Structural Robustness and Reconfiguration of Global Crude Oil Trade in 2017, 2019, and 2020

Dear Editor and Reviewers,

Thank you for evaluating the original manuscript, Structural Resilience of the Global Crude Oil Trade Network under Systemic Crises and Sanctions and Implications for Energy Security and Sustainable Transitions. Reviewer 1 recommended acceptance. The revised manuscript responds to the Editor and the remaining reviewers through a new, directly measured robustness analysis and a full reconstruction of the directed and monetary network definitions. The observation years remain 2017, 2019 and 2020.

All numerical changes reported below come from the authors' consolidated datasets and the documented scripts. A newly received 2019 reporter-level archive was checked against the 2019 consolidated network; discrepancies are disclosed openly, not replaced with invented values. The repository folder has been prepared for publication under ResearchRepro, but no claim is made that the GitHub repository is already online.

REVISION CONVENTION

The complete revised body is blue-marked in the review version. The clean manuscript is provided separately for submission. Bold type in this response letter emphasizes approximately one-fifth of response text. Locations refer to stable section, table and figure numbers instead of pages.

Revision at a Glance

Revision area	Action taken	Location
Design and claims	Retained 2017/2019/2020; no new years or causal inference	Abstract; Sections 1, 3.4, 6–7
Data provenance	Added OEC HS 2709 processing and 2019 archive reconciliation audit	Section 3.1; Table 2; GitHub audit files
Directed weighted network	Formalized adjacency, in/out degree and USD strengths	Sections 3.2–3.3; Tables 3–5
Centrality	Added weighted HITS and outward harmonic measures	Sections 3.3 and 4.2; Tables 4–5
Community structure	Weighted Louvain projection, fixed resolution and seed checks	Section 4.3; Table 7
Robustness and sensitivity	Deletion tests, efficiency, threshold sensitivity and null tests	Section 4.4; Tables 9–11; Figures 7–8
Reproducibility	Prepared public repository, code,	Declarations; research package README

Revised manuscript: full replacement text indicated in the blue-marked version.

	cleaned edges, GEXF and tables	
Manuscript structure	11 tables, 8 figures, EndNote fields, live SEQ/REF cross-references	Entire manuscript; EndNote import RIS

Point-by-point response to the Editor

Editor Comment 1

REVIEWER / EDITOR COMMENT

Due the comments from the reviewer's, I suggest a major revision for this manuscript.

RESPONSE

The manuscript has been rewritten with directed and dollar-weighted measures, corrected reachability analysis, community detection specifications and independent network stress tests. Claims beyond the fixed three-year record have been removed.

LOCATION IN REVISED MANUSCRIPT New manuscript, especially Sections 3–7; Tables 3–11; Figures 3–8.

Editor Comment 2

REVIEWER / EDITOR COMMENT

Declaration Section: Authors are requested to include a separate section in the manuscript titled "Declarations." In this section, please clearly state Ethical approval, Consent to participate, and Consent to publish as individual subheadings.

RESPONSE

A separate Declarations section contains each requested heading and a study-specific statement. Funding, competing interests, data/code availability, author-contribution verification and AI-assisted preparation are included as distinct items.

LOCATION IN REVISED MANUSCRIPT Declarations section immediately before References.

Point-by-point response to Reviewer 1

Reviewer 1 Comment 1

REVIEWER / EDITOR COMMENT

The author has made careful revisions in accordance with the reviewers' comments and the manuscript has now reached the publication standard. It is recommended for acceptance.

RESPONSE

The authors appreciate the favorable assessment and the recommendation for acceptance. The present revision addresses requests from the Editor and Reviewers 2 and 3 without adding new observation years or altering the underlying research question beyond its measurable scope.

LOCATION IN REVISED MANUSCRIPT Title and scope of the revised manuscript; Sections 1 and 7.

Point-by-point response to Reviewer 2

Reviewer 2 Comment 1

REVIEWER / EDITOR COMMENT

Revised manuscript: full replacement text indicated in the blue-marked version.

Authors actually analyze changes in the structure of the oil trade network comparing tree points in time, therefore, their contribution is limited to that. Authors do not measure resilience nor effects of systemic crises. Therefore, Authors should change the title and presentation of their paper accordingly.

RESPONSE

The title, Abstract and research aims have been rewritten to describe three annual network snapshots. Simulated node-removal robustness is now measured directly, while no causal or realized-recovery effect is claimed.

LOCATION IN REVISED MANUSCRIPT Title; Abstract; Sections 1, 3.4, 4.4, 6–7; Table 9; Figure 7.

Reviewer 2 Comment 2

REVIEWER / EDITOR COMMENT

Authors should specify all procedures performed for their “data cleaning” listed in Figure 1

RESPONSE

The processing steps are now explicit: retain positive numeric weighted edges, remove self-links, standardize a small set of demonstrably equivalent jurisdiction labels, and aggregate duplicate ordered pairs. The original labels, rows, changes and a newly supplied 2019 source crosswalk are retained. Because many source values do not match the consolidated network, the uncertainty is disclosed rather than silently harmonized.

LOCATION IN REVISED MANUSCRIPT Sections 3.1–3.2; Table 2; data/audit/data_cleaning_audit.csv; raw_2019_crosswalk.csv.

Reviewer 2 Comment 3

REVIEWER / EDITOR COMMENT

Present formally the definition of the networks and precise if networks are binary or weighted. As it is stated it is not clear.

RESPONSE

The adjacency indicator and monetary weight are formally defined. Each ordered edge records source-to-destination trade, and all monetary weights are United States dollars, not millions. Binary measures and dollar-weighted measures are explicitly distinguished.

LOCATION IN REVISED MANUSCRIPT Section 3.2; equations defining A and w; Section 3.3.

Reviewer 2 Comment 4

REVIEWER / EDITOR COMMENT

Reading “Social Network Analysis Metrics” one would understand that the network is undirected. But, authors present their research question as the analysis of the oil trade network, that is exports and imports, therefore, the network should be directed. If it is no directed explain and justify why. If it is directed, then stated this more clearly in the text.

RESPONSE

The primary networks are directed. Both reciprocal directions are retained if observed; the definition, density denominator, incoming and outgoing roles, and strong-versus-weak component

distinction are provided. An undirected projection is used only for declared clustering, community detection and weak-connectivity stress testing.

LOCATION IN REVISED MANUSCRIPT Sections 3.2–3.4; Tables 3–4; Figures 3–4.

Reviewer 2 Comment 5

REVIEWER / EDITOR COMMENT

1.3 Average degree is not informative. The Degree distribution of many economic networks is highly skewed, therefore, an average is not informative. Authors should present degree distributions with best fits and observe if these best fits change from one year to another.

RESPONSE

Separate positive in-degree and out-degree distributions are visualized. Geometric and Zipf candidate distributions are fitted and their information criteria compared; the paper does not claim an established universal power-law fit.

LOCATION IN REVISED MANUSCRIPT Section 3.4; Section 4.2; Table 6; Figure 5; supplementary/degree_distribution_model_fit.csv.

Reviewer 2 Comment 6

REVIEWER / EDITOR COMMENT

Since authors are analyzing the oil trade network, they should consider reporting degree according to directionality. Therefore, authors should report in degree centrality and out degree centrality.

RESPONSE

Country-by-year in-degree, out-degree, in-strength and out-strength are all computed. The results distinguish breadth of export partners from monetary export values and identify leaders by each measure.

LOCATION IN REVISED MANUSCRIPT Section 3.2; Section 4.2; Table 4; supplementary/centrality_2017–2020.csv.

Reviewer 2 Comment 7

REVIEWER / EDITOR COMMENT

Authors should provide an economic interpretation of betweenness centrality, which is not provided.

RESPONSE

Betweenness is interpreted as participation in hypothetical directed topological shortest paths. It is explicitly not treated as physical oil transit, control of shipments, or proof of commercial intermediation.

LOCATION IN REVISED MANUSCRIPT Sections 3.3 and 5.1.

Reviewer 2 Comment 8

REVIEWER / EDITOR COMMENT

Authors should include an eigenvalue-based centrality such as authority and hub scores to fully characterize the structure of the oil trade network.

RESPONSE

Directed value-weighted HITS hub and authority measures have been added and reported alongside directional degree and strength for every country. The original unmeasured centers-of-influence interpretation has been removed.

LOCATION IN REVISED MANUSCRIPT Section 3.3; Section 4.2; Table 4; three centrality CSVs.

Reviewer 2 Comment 9

REVIEWER / EDITOR COMMENT

Figures 3, 4, and 5 are too small.

RESPONSE

The three earlier undersized network panels have been replaced by a large common-layout triptych. All scientific figures have high-resolution PNG and editable SVG versions.

LOCATION IN REVISED MANUSCRIPT Figure 3; figures/Figure_03_*.png and .svg.

Reviewer 2 Comment 10

REVIEWER / EDITOR COMMENT

Provide association between rankings of network measures for the tree years to observe how similar-different are from one year to another.

RESPONSE

Pairwise Spearman coefficients are reported on the country-label intersection for each pair of snapshots. In-degree, out-degree, value strength and harmonic-centrality ranks are compared.

LOCATION IN REVISED MANUSCRIPT Section 4.3; Table 8; supplementary/rank_stability.csv.

Reviewer 2 Comment 11

REVIEWER / EDITOR COMMENT

discussion in section 5 should be in Capital letter as Discussion

RESPONSE

The section heading has been normalized to Discussion with consistent numbering and capitalization.

LOCATION IN REVISED MANUSCRIPT Section 5.

Reviewer 2 Comment 12

REVIEWER / EDITOR COMMENT

Discussion section should be completely written in paragraphs not bullets.

RESPONSE

The Discussion has been rewritten in connected paragraphs. The old list-like question-by-question structure has been removed.

LOCATION IN REVISED MANUSCRIPT Sections 5.1–5.3.

Reviewer 2 Comment 13

REVIEWER / EDITOR COMMENT

Tune down interpretations and conclusions. Authors do not provide evidence about resilience nor

about the effect of systemic crises. Author should limit their interpretation to the reach of the network measures they report to characterize the structure of the oil trade network and changes comparing tree years.

RESPONSE

The interpretation now separates observed cross-year changes from hypothetical network robustness. Sanctions and COVID-19 are historical contexts; direct causal attribution is not asserted.

LOCATION IN REVISED MANUSCRIPT Abstract; Sections 1, 5–7.

Reviewer 2 Comment 14

REVIEWER / EDITOR COMMENT

In general, points in the discussion face two problems: on one hand the interpretations provided are not completely based on results but go far beyond and are not evidence-based. On the other hand, they stay too superficial. In particular, Point 1 in discussion is reporting a formal result. But what about the intermediary role that Iran maintains informally? Interpretation cannot be “to the letter”.

RESPONSE

The interpretation of Iran is restricted to its reported formal trade relationships and graph-derived centralities. Informal or concealed oil trade is explicitly outside the observations and cannot be reconstructed from these files.

LOCATION IN REVISED MANUSCRIPT Sections 4.2 and 5.1; Table 5.

Reviewer 2 Comment 15

REVIEWER / EDITOR COMMENT

Point 2 in discussion is not based on results from the analysis. Authors cannot know this if they are not analyzing a directed network. More over, authors are not analyzing petroleum demand, or export volumes and revenues, so cannot interpret in those terms.

RESPONSE

The directed out-degree and out-strength now make export-side claims measurable. The paper does not claim physical volume, demand or price-independent income effects, since the inputs contain nominal dollar values only.

LOCATION IN REVISED MANUSCRIPT Sections 3.2, 4.2, 5.1 and 6; Tables 4–5.

Reviewer 2 Comment 16

REVIEWER / EDITOR COMMENT

In relation to point 3 of the discussion, authors are not analyzing or measuring resilience therefore, cannot interpret in those terms.

RESPONSE

Simulated robustness is directly assessed through targeted and randomized node deletion, together with directed efficiency and component statistics. These measures are carefully distinguished from realized recovery and supply replacement.

LOCATION IN REVISED MANUSCRIPT Sections 3.3–3.4, 4.4 and 6; Tables 3 and 9; Figure 7.

Revised manuscript: full replacement text indicated in the blue-marked version.

Reviewer 2 Comment 17

REVIEWER / EDITOR COMMENT

in point 4 authors are interpreting in terms of “centers of influence” but this is not measured. None of the centrality measures included in the analysis are measuring centers of influence. To measure influence, authors should include a centrality measure based on eigenvalue.

RESPONSE

The label centers of influence was removed as an untested substantive claim. Weighted HITS authorities and hubs now document an eigenvalue-based network position, without implying political influence.

LOCATION IN REVISED MANUSCRIPT Section 3.3; Section 4.2; Table 4.

Reviewer 2 Comment 18

REVIEWER / EDITOR COMMENT

Revise conclusion section according to comments. In particular, tune down the reach of the analysis. It was not detailed, resilience was not measured, with tree years one cannot talk about evolution.

RESPONSE

The Conclusion was rewritten around measured snapshot differences and bounded artificial-disruption outcomes. No long-run trajectory or sanctions/pandemic treatment effect is inferred from three years.

LOCATION IN REVISED MANUSCRIPT Section 7.

Reviewer 2 Comment 19

REVIEWER / EDITOR COMMENT

For transparency and reproducibility, all data and code should be fully available for anyone. Not on reasonable request. Provide data and code using a repository such as git hub or Open Science Framework.

RESPONSE

A standalone, documented GitHub-ready package contains cleaned directed edge lists, reproducible code, numerical outputs, GEXF networks and visualization assets. It is prepared for upload to ResearchRepro; public availability has not yet been asserted. Raw OEC file redistribution permissions require author verification.

LOCATION IN REVISED MANUSCRIPT Declarations, Data and code availability; GitHub-ready README; LICENSE_CODE_MIT and DATA_REUSE_NOTICE.md.

Point-by-point response to Reviewer 3

Reviewer 3 Comment 1

REVIEWER / EDITOR COMMENT

The paper should be more careful in describing the effects of sanctions and COVID-19. The analysis only compares 2017, 2019, and 2020. These three years can show changes before and after the two events, but they are not sufficient to identify causal effects. Other factors also changed during this

period. The authors should reduce causal wording throughout the paper and use expressions such as "associated with" or "during the post-sanctions period." If possible, adding more consecutive years would make the comparison more convincing.

RESPONSE

The three historical snapshots are preserved deliberately because expanding the study years would redefine the project. All causal formulations have been removed. The limitations explicitly identify the lack of a counterfactual and the simultaneous influence of other oil-market changes.

LOCATION IN REVISED MANUSCRIPT Abstract; Sections 1, 3.4, 5.1, 6–7.

Reviewer 3 Comment 2

REVIEWER / EDITOR COMMENT

The construction of the trade network is not sufficiently clear. The paper states that an edge exists when two countries have an import or export relationship, but it is not clear whether the final network is directed or undirected, how reciprocal trade is treated, and whether import and export records are combined. These details directly affect density, degree, betweenness, closeness, and the number of components. The revised paper should clearly define the adjacency matrix, edge direction, weights, and data-cleaning rules.

RESPONSE

The directed adjacency and weight definitions, treatment of reciprocal relations, cleaning rules, name audit and weak/strong component conventions are now stated. The 2019 reporter-level source audit exposes unresolved monetary discrepancies instead of treating a merged importer/exporter value as unambiguously verified.

LOCATION IN REVISED MANUSCRIPT Sections 3.1–3.3; Table 2; data/audit/.

Reviewer 3 Comment 3

REVIEWER / EDITOR COMMENT

The paper uses trade-value data, but most of the analysis appears to be based on binary links. This means that a small trade relationship and a very large trade relationship may be treated in the same way. For crude oil trade, this is an important limitation. The authors should add weighted network measures, such as strength, weighted clustering, or weighted betweenness, or provide a clear reason for using only binary networks. A comparison between weighted and unweighted results would also improve the paper.

RESPONSE

The analysis now reports dollar-valued in/out strengths, value-weighted HITS, weighted Louvain modularity and a declared undirected weighted clustering calculation. Unweighted directed betweenness remains complementary because dollar values cannot be interpreted as path lengths without a substantive cost transformation.

LOCATION IN REVISED MANUSCRIPT Sections 3.2–3.3 and 4.2–4.3; Tables 4 and 7; country-level CSV files.

Reviewer 3 Comment 4

REVIEWER / EDITOR COMMENT

Revised manuscript: full replacement text indicated in the blue-marked version.

The closeness-centrality results need to be checked carefully. The network contains a very large number of components, but Table 5 reports a closeness centrality of 1.0 for Niger and Togo in 2020, which is higher than the value for the United States. This result is difficult to interpret in a highly disconnected network and may reflect the way closeness is calculated in Gephi. The authors should explain the exact definition used and reconsider whether harmonic centrality or closeness within the largest connected component would be more appropriate.

RESPONSE

The previous closeness ranking, including the anomalous values of one for Niger and Togo, was removed. Normalized outward harmonic centrality now treats unreachable destinations as zero and is computed for all countries and years.

LOCATION IN REVISED MANUSCRIPT Section 3.3; Table 5; supplementary/centrality_*.csv.

Reviewer 3 Comment 5

REVIEWER / EDITOR COMMENT

The paper repeatedly uses the term “resilience,” but resilience is not directly measured. Density, degree, clustering, centrality, and modularity describe network structure, but they are not direct measures of network resilience. The decline in these indicators does not automatically mean that the network became less resilient. The authors should either narrow the terminology to structural change or network reconfiguration, or add direct resilience measures, such as global efficiency, largest connected component, node-removal tests, or network robustness under targeted shocks.

RESPONSE

The manuscript now operationalizes simulated structural robustness with direct node-deletion experiments and directed global efficiency, and presents strong/weak component statistics. Recovery time, physical redistribution and actual oil security are not claimed.

LOCATION IN REVISED MANUSCRIPT Sections 3.3–3.4, 4.1, 4.4 and 6; Tables 3 and 9; Figure 7.

Reviewer 3 Comment 6

REVIEWER / EDITOR COMMENT

The modularity analysis needs more detail and more cautious interpretation. The paper reports different numbers of communities in 2017, 2019, and 2020, but the algorithm settings, modularity scores, resolution parameter, and stability of the results are not fully reported. The names assigned to the communities also appear to be subjective. The revised version should provide the full community-detection procedure and use more neutral labels unless the economic or regional interpretation can be supported more clearly.

RESPONSE

Community detection is fully specified as weighted Louvain on the undirected sum-of-reciprocals projection, resolution one, displayed seed 46 and twenty alternate seeds. Community names are neutral identifiers and the seed-dependent ranges are presented.

LOCATION IN REVISED MANUSCRIPT Sections 3.3 and 4.3; Table 7; supplementary/community_seed_stability.csv.

Reviewer 3 Comment 7

REVIEWER / EDITOR COMMENT

Some interpretations go beyond the reported evidence. For example, the paper links the strong position of the United States to economic diversity and strategic adaptability, and argues that central and large countries were more resilient. These explanations are plausible, but they are not directly tested in the analysis. The discussion should separate what is directly shown by the network results from possible explanations. If the paper wants to make claims about central versus peripheral or large versus small countries, additional group comparisons are needed.

RESPONSE

Unsupported explanations based on national diversity, strategic adaptability, geopolitical power and physical demand have been removed. All remaining comparisons concern measured directional degree, strength, HITS, harmonic centrality or graph connectivity.

LOCATION IN REVISED MANUSCRIPT Sections 4.2, 5.1 and 6; Tables 4–5.

Reviewer 3 Comment 8

REVIEWER / EDITOR COMMENT

The paper needs at least some robustness or significance checks. At present, most conclusions are based on direct comparisons of network statistics across three years. It is not clear whether the observed changes are large enough to be considered meaningful or whether they are sensitive to network construction. The authors should add robustness tests, for example using alternative trade-value thresholds, weighted versus unweighted networks, a common-country sample, permutation tests, or null-network comparisons.

RESPONSE

Sensitivity checks now vary three dollar-value cutoffs, compare ranks for common country sets, fit degree distributions, and use 25 degree-preserving rewiring benchmarks. The paper explains that these tests describe construction sensitivity rather than causal significance.

LOCATION IN REVISED MANUSCRIPT Sections 3.4 and 4.2–4.4; Tables 6, 8, 10 and 11; Figures 5 and 8.

Reviewer 3 Comment 9

REVIEWER / EDITOR COMMENT

The discussion of the SDGs should be separated more clearly from the empirical findings. The paper itself notes that it does not directly measure official SDG indicators. Therefore, the links to SDGs 7, 9, 13, and 17 should be presented as policy implications rather than direct empirical results. This is particularly important for claims related to decarbonization and climate action, which are not directly tested in the current analysis.

RESPONSE

SDGs 7, 9, 13 and 17 are discussed only as bounded policy considerations. No SDG indicator, emissions effect or decarbonization outcome is reported as an empirical result.

LOCATION IN REVISED MANUSCRIPT Section 5.3 and Section 6.

Reviewer 3 Comment 10

REVIEWER / EDITOR COMMENT

Revised manuscript: full replacement text indicated in the blue-marked version.

The paper still needs a careful technical revision. There are inconsistencies in section numbering, table structure, year ordering, terminology, and capitalization. The Results section begins as Section 4, but the following subsections are numbered 5.1, 5.2, and so on. Some parts of the text also read like editing remnants rather than final manuscript language. The authors should carefully check all section numbers, tables, figures, variable definitions, and numerical descriptions before resubmission.

RESPONSE

The manuscript was restructured with sequential section numbering, uniform terms and year ordering, consistent figure and table captions, and Word SEQ/REF cross-reference fields. The abstract has fewer than 200 words and exactly five keywords.

LOCATION IN REVISED MANUSCRIPT Entire clean manuscript; eleven tables and eight figures; Word field audit.

Outstanding author verification before upload

The following items cannot be completed accurately without author confirmation: the identity of the 2019 source label Chinese, the historical cause of dollar-value differences between 2019 files, the scope of permission to redistribute OEC-derived values, the actual publication status of the GitHub repository, the author-contribution statement, and the EndNote Update Citations and Bibliography test in the authors' EndNote installation. These items are disclosed rather than recorded as already resolved.

The revised manuscript and analytical repository package have been produced for author validation. Thank you for the specific methodological observations, which led to a more constrained and reproducible account of the original three-year comparison.